

Ayush Chauhan

github.com/bakayu

mail@ayuch.dev | +91 8448522434
ayuch.dev/blog | linkedin.com/in/ayuch

Education

Chitkara University

Bachelor of Engineering in Computer Science and Engineering; CGPA: 9.31

- Coursework: Data Structures and Algorithms, OS, Computer Networks, Distributed Systems & Software Design

Rajpura, Punjab, India

Expected Aug. 2028

Experience

Google Summer of Code – OpenPrinting/System76 | Software Engineering Intern May 2026 – Aug. 2026

- Fixed a missing shared-library link dependency in cpdb-libs (C), resolving a broken libtool LIBADD reference in libcpdb-frontend.so - merged upstream
- Ported C-based CUPS utilities to Rust to validate new FFI bindings, exercising edge cases across the C/Rust boundary in backend discovery and print-job submission
- Integrated zbus into cpdb-rs, replacing glib-based IPC with asynchronous D-Bus support and enabling compatibility with pure async Rust environments
- Authored 5+ code examples illustrating async D-Bus integration and CPDB workflows for Rust applications
- Built a CPDB-based print dialog frontend for the COSMIC desktop in Rust and libcosmic, providing native printing workflows for Linux desktop and Flatpak applications

LF Decentralized Trust | Open Source Mentee

Jun. 2025 – Nov. 2025

- Ported an MCP NTP Timeserver to a TMCP service, enabling trusted time verification for AI agent ecosystems
- Designed 2 JSON-RPC protocols for credential issuance and presentation with replay protection, selective disclosure, and P-256 cryptographic operations, and validated interoperability with open-source LLMs including Qwen and Llama

Projects

Raven | Rust, Tokio, Tonic, Axum, ClickHouse, VictoriaMetrics | github.com/rvnhq/raven

- Engineered a Rust observability platform with a Linux agent, gRPC ingestion service, and control plane; agent performs inotify-based log tailing and procfs sampling for low-overhead system monitoring
- Designed a write-ahead-log (WAL) buffering layer for crash-safe telemetry delivery under network partition, with TLS-secured transport between agent and ingestion service
- Sustained 41K+ HTTP requests/sec and 900+ gRPC RPCs/sec on a single container instance during concurrent load testing

Gitx | Go, Bubbletea, GitHub Actions | github.com/gitxtui/gitx

- Created a terminal-based Git client in Go supporting staging, commit history inspection, branching, stash management, and repository navigation
- Distributed cross-platform binaries for Linux, macOS, and Windows, reaching 200+ downloads while coordinating contributions from 8 developers

pack | C, POSIX | github.com/bakayu/pack

- Built a recursive file/directory archiver in C with a Huffman compression layer built from scratch (no external libraries), reaching upto 40% size reduction on source trees
- Designed a versioned binary container which includes a magic header, pre-order-serialized code tree, bit-packed body, with a custom bit level reader/writer min-heap, and preserved POSIX dirent/stat permission bits

Community and Open Source

Technical Lead & Core Team | Open Source Chandigarh

Sep. 2024 – Present

- Leading technical direction and architecture for community projects; Delivered Git, GitHub, and Linux workshops introducing 150+ participants to command-line tooling and open-source development practices

Open Source Contributor | Graphite, OpenWallet Foundation, OpenPrinting, Layer5io

Aug. 2024 – Present

- Contributed to open-source organizations with work spanning procedural 2D vector graphics, developer tooling, identity infrastructure, and Linux printing subsystems; maintained public CLI tools and backend libraries

Technical Skills

Languages: Rust, C, Golang, Java, Python, TypeScript, SQL (PostgreSQL)

Systems and Backend: Tokio, async runtimes, D-Bus (zbus), IPC, inotify, Axum, Actix-web, Tonic (gRPC), Gin

DevOps and Tools: Linux, Bash, Git, Docker, Docker Compose, GitHub Actions, Kubernetes, Azure